

MULTI-MAP FUSION FOR WEAKLY SUPERVISED DISEASE LOCALIZATION FROM GLOBALLY ASSIGNED DIAGNOSTIC LABELS IN BRAIN MRI

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ABSTRACT

Deep learning for brain MRI analysis is often limited by the need for dense voxelwise annotations, which are costly and unreliable for subtle abnormalities. We propose GALA-Former, a weakly supervised Transformer-based framework that localizes disease-relevant regions using only subject-level diagnostic labels propagated to 2D slices. The method aggregates complementary internal localization signals—token-level activation, attention rollout, and gradient-based relevance—through a confidence-guided fusion module, enabling anatomically coherent and interpretable localization without spatial supervision. Experiments on three T₁-weighted MRI datasets spanning tumors, multiple sclerosis, and Alzheimer’s disease demonstrate robust generalization and consistently improved localization over existing weakly supervised approaches.

Index Terms— Weakly supervised learning, Transformer, Anomaly localization, Interpretability, Clinical MRI

1. INTRODUCTION

Deep learning has substantially advanced brain Magnetic Resonance Imaging (MRI) analysis, particularly through Transformer architectures with strong global modeling capabilities [1]. However, clinical deployment remains limited by the need for *dense voxelwise segmentation masks*—3D annotations specifying pathology at each voxel—which are costly, time-consuming, and often unreliable for subtle abnormalities such as small vascular lesions or diffuse white-matter degeneration [2, 3, 4]. This motivates localization methods that rely only on readily available diagnostic labels.

To this end, we introduce **GALA-Former**, a Gradient-, Attention-, and token-wise Localization Aggregation TransFormer for fine-grained anomaly localization using only *global subject-level labels*. The model aggregates complementary *internal localization signals*—token-level activation, attention rollout, and gradient-based relevance—and fuses them via a lightweight Convolutional Neural Network (CNN) to generate anatomically coherent localization maps without spatial supervision.

Existing weak-supervision strategies have notable limitations: Class Activation Map (CAM)-based approaches produce coarse maps [5], pseudo-label refinement accumulates errors [6], and single-cue Transformer interpretations remain insufficient for anatomically faithful localization [7]. These gaps motivate a *multi-cue Transformer-based* framework. Unlike existing approaches that rely on a single attribution source or require additional spatial or instance-level supervision, GALA-Former jointly exploits multiple complementary internal signals under purely diagnostic-label supervision, enabling stable and anatomically meaningful localization across heterogeneous MRI conditions.

We evaluate GALA-Former on three T₁-weighted MRI datasets spanning tumors (BraTS) [8], multiple sclerosis (MSLesSeg) [9], and Alzheimer’s disease (ADNI) [10]. The method delivers more robust, interpretable, and transferable localization than existing weakly supervised approaches.

Our contributions are:

- A Transformer-based weakly supervised framework for pathology localization using only global diagnostic labels.
- A multi-map fusion strategy that combines complementary token-level activation, attention-derived, and gradient-based localization signals at the token level for anatomically consistent localization.
- Demonstrated generalization across heterogeneous MRI cohorts without voxelwise spatial supervision.

2. WEAKLY SUPERVISED ANOMALY LOCALIZATION

2.1. Slice-wise Transformer Representations

CNN-based classifiers are limited in capturing long-range contextual patterns. Vision Transformers (ViT) [7, 11] address this limitation through global self-attention, while hybrid medical architectures [12, 13] improve spatial reasoning but still rely on voxel-level annotations, limiting scalability under diagnostic-label supervision. Self-supervised Transformer models [14] preserve patch-level structure via distillation, yielding dense features suitable for weak localization.

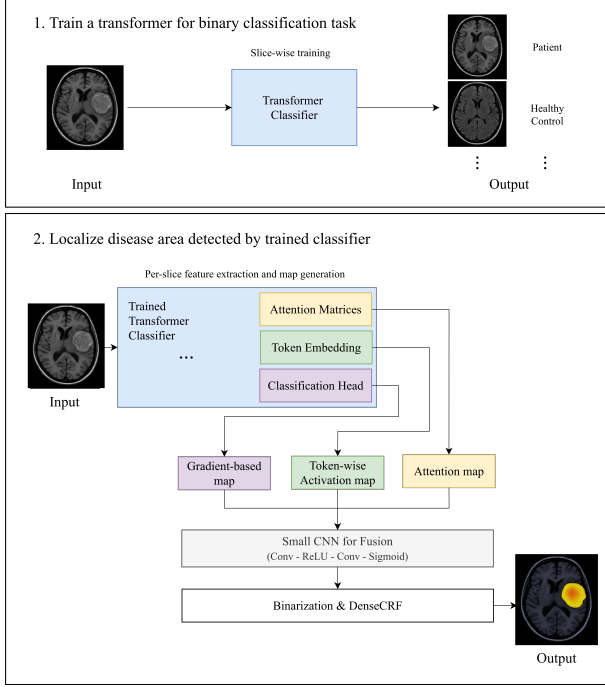


Fig. 1. Overview of the proposed GALA-Former framework: Top, slice-wise classification using subject-level labels propagated to 2D slices; Bottom, fusion of token activation, attention rollout, and gradient relevance for localization.

2.2. Weak Localization from Diagnostic Labels

Weakly supervised localization (WSL) infers spatial relevance from global labels. CAM methods [15, 16, 17] highlight only the most discriminative regions, while Transformer-based WSL often relies on a single cue, limiting robustness in diffuse or multi-focal abnormalities [18]. Recent Transformer-based methods assume multiple-instance or region-level supervision and are not applicable to purely diagnostic-label settings.

GALA-Former aggregates complementary internal localization signals—token-level activation, attention rollout, and gradient-based relevance—through a confidence-guided fusion CNN, enabling anatomically coherent localization using only diagnostic labels.

3. THE GALA-FORMER FRAMEWORK

We propose a weakly supervised anomaly localization framework trained solely with globally assigned diagnostic labels propagated to 2D slices (Fig. 1). Each slice is processed independently, without voxelwise annotations.

3.1. Feature Extraction and Localization Cues

Given an image slice $I \in \mathbb{R}^{H \times W}$, a pretrained Transformer classifier f_θ divides it into N non-overlapping patches, each

mapped to a D -dimensional embedding. Let $t_i \in \mathbb{R}^D$ denote the patch embedding for patch i and t_{cls} the learnable class token. The Transformer encoder outputs the final class token, which is passed through a linear classification head f_{head} :

$$\mathbf{y} = f_{\text{head}}(\text{Transformer}(\{t_i\} \cup t_{\text{cls}})), \quad (1)$$

where $\mathbf{y} \in \mathbb{R}^C$ contains logits over C diagnostic classes.

We extract three complementary localization cues from the trained classifier:

(i) Token-wise Map Patch relevance is computed via cosine similarity between each patch token t_i and the class token t_{cls} :

$$A_{\text{token}}(i) = \frac{t_i^\top t_{\text{cls}}}{\|t_i\|_2 \|t_{\text{cls}}\|_2} \quad (2)$$

(ii) Attention Map For each Transformer layer $\ell \in \{1, \dots, L\}$, let $\mathbf{W}^{(\ell)} \in \mathbb{R}^{(N+1) \times (N+1)}$ denote the attention matrix including the class token. Attention rollout aggregates attention through sequential matrix multiplication:

$$A_{\text{attn}} = \frac{1}{L} \sum_{\ell=1}^L \left(\prod_{k=1}^{\ell} \mathbf{W}^{(k)} \right) \quad (3)$$

(iii) Gradient Map A gradient map is derived by computing gradients with respect to the pre-softmax logit y_c to avoid softmax saturation. Let c denote the predicted class index (argmax of the logit vector $\mathbf{y}$), and $F_k(x, y)$ the activation at spatial location (x, y) of channel k in the final feature map before pooling:

$$A_{\text{grad}}(x, y) = \text{ReLU} \left(\sum_k \frac{\partial y_c}{\partial F_k(x, y)} F_k(x, y) \right) \quad (4)$$

where $\text{ReLU}(z) = \max(z, 0)$ is the Rectified Linear Unit [19]. All maps A_m ($m \in \{\text{token}, \text{attn}, \text{grad}\}$) are upsampled to $H \times W$ and normalized to $[0, 1]$.

3.2. Fusion and Refinement

The three cues are stacked channel-wise:

$$\mathbf{A}_{\text{in}} = [A_{\text{token}}, A_{\text{attn}}, A_{\text{grad}}] \in \mathbb{R}^{H \times W \times 3}. \quad (5)$$

A lightweight fusion module with two convolutional layers produces the combined map:

$$A_{\text{fused}} = \sigma(\text{Conv}_2(\text{ReLU}(\text{Conv}_1(\mathbf{A}_{\text{in}})))) \quad (6)$$

where Conv_1 is a 3×3 convolution, Conv_2 a 1×1 projection, and σ the sigmoid function.

The final localization map is obtained by binary thresholding followed by DenseCRF refinement [20], which enforces spatial coherence and sharpens boundaries via spatial- and intensity-based pairwise potentials. DenseCRF is applied with spatial kernel = 3, bilateral kernel = 10, compatibility coefficient = 3, and 10 mean-field inference iterations.

Table 1. Slice-level classification results with 95% confidence intervals on BraTS, MSLesSeg, and ADNI.

DATASET	METHOD	AUROC	AUPRC	ACCURACY	F1 SCORE
BRATS [8]	RESNET-50 [21]	0.94 ± 0.02	0.90 ± 0.01	0.91 ± 0.02	0.89 ± 0.02
	ViT [7]	0.97 ± 0.01	0.94 ± 0.005	0.94 ± 0.01	0.93 ± 0.01
	DINOv3 [14]	0.98 ± 0.01	0.96 ± 0.004	0.96 ± 0.01	0.95 ± 0.01
MSLESSEG [9]	RESNET-50 [21]	0.89 ± 0.03	0.72 ± 0.02	0.84 ± 0.03	0.80 ± 0.03
	ViT [7]	0.92 ± 0.02	0.78 ± 0.01	0.87 ± 0.02	0.85 ± 0.02
	DINOv3 [14]	0.94 ± 0.02	0.82 ± 0.01	0.89 ± 0.02	0.88 ± 0.02
ADNI [10]	RESNET-50 [21]	0.76 ± 0.04	0.61 ± 0.03	0.70 ± 0.04	0.68 ± 0.04
	ViT [7]	0.80 ± 0.03	0.66 ± 0.02	0.74 ± 0.03	0.72 ± 0.03
	DINOv3 [14]	0.84 ± 0.03	0.71 ± 0.02	0.78 ± 0.03	0.76 ± 0.03

4. EXPERIMENTAL SETUP

4.1. Datasets and Preprocessing

We use T_1 -weighted MRI from BraTS [8], MSLesSeg [9], and ADNI [10]. All datasets were normalized to MNI space (1 mm isotropic)¹, skull-stripped, bias-field corrected, and re-sampled into 224×224 axial slices.

BraTS Gadolinium-enhanced T_1 -weighted images provide clearly visible tumors. From 369 subjects (42k slices), lesion-free slices were labeled as *control* and lesion-containing slices as *patient*; 18k slices with masks were used for localization.

MSLesSeg Multiple sclerosis lesions typically appear as T_1 -weighted hypointensities. Data from 50 subjects yielded 9k slices for classification and 4k slices with manual segmentations for localization.

ADNI Alzheimer’s disease is not visually detectable on T_1 -weighted MRI. Volumes (typically $170 \times 256 \times 256$) were used only for classification and unsupervised localization. Subject-level diagnoses—Cognitively Normal (CN) vs. Alzheimer’s Disease (AD)—were uniformly propagated to slices, producing 120k slice-level samples from 710 subjects.

4.2. Backbones and Training

We evaluate ResNet-50 [21], ViT-B/16 [7], and DINOv3 [14], with corresponding localization variants: (1) ResNet-50 + Grad-CAM [16], (2) ViT/DINOv3 with attention rollout, (3) ViT/DINOv3 with Grad-CAM.

Models were trained for 100 epochs using AdamW [22] (learning rate 10^{-4} , weight decay 10^{-2} , batch size 32) with standard augmentations and cosine decay. Each experiment was repeated five times; results are reported as mean and 95% confidence intervals. Classification was evaluated strictly at the slice level.

4.3. Evaluation Metrics

Classification We report Area Under the Receiver Operating Characteristic Curve (AUROC) / Area Under the Precision-Recall Curve (AUPRC), which are robust to class imbalance, together with Accuracy and the maximum F1-score across thresholds.

¹SPM12: <https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>

Table 2. Localization performance with 95% confidence intervals on the BraTS dataset.

METHOD	DICE	mIoU	L-REC	L-PREC	MAP@0.5
RESNET-50 [21] + GRAD-CAM [16]	0.49 ± 0.02	0.35 ± 0.01	0.54 ± 0.02	0.40 ± 0.03	0.37 ± 0.02
ViT [7] ATTENTION ROLLOUT	0.50 ± 0.03	0.42 ± 0.03	0.61 ± 0.01	0.45 ± 0.01	0.43 ± 0.01
ViT [7] + GRAD-CAM [16]	0.62 ± 0.01	0.49 ± 0.01	0.65 ± 0.04	0.55 ± 0.01	0.48 ± 0.03
DINOv3 [14] ATTENTION ROLLOUT	0.59 ± 0.02	0.45 ± 0.02	0.64 ± 0.01	0.50 ± 0.02	0.47 ± 0.01
DINOv3 [14] + GRAD-CAM [16]	0.67 ± 0.01	0.54 ± 0.01	0.72 ± 0.02	0.63 ± 0.01	0.56 ± 0.01
GALA-FORMER	0.71 ± 0.03	0.58 ± 0.03	0.84 ± 0.03	0.68 ± 0.01	0.61 ± 0.02

Table 3. Localization performance with 95% confidence intervals on the MSLesSeg dataset.

METHOD	DICE	mIoU	L-REC	L-PREC	MAP@0.5
RESNET-50 [21] + GRAD-CAM [16]	0.46 ± 0.03	0.33 ± 0.02	0.52 ± 0.01	0.39 ± 0.03	0.35 ± 0.04
ViT [7] ATTENTION ROLLOUT	0.52 ± 0.02	0.37 ± 0.01	0.58 ± 0.02	0.42 ± 0.01	0.40 ± 0.03
ViT [7] + GRAD-CAM [16]	0.60 ± 0.02	0.47 ± 0.03	0.63 ± 0.01	0.56 ± 0.02	0.49 ± 0.02
DINOv3 [14] ATTENTION ROLLOUT	0.56 ± 0.02	0.43 ± 0.01	0.61 ± 0.02	0.48 ± 0.03	0.44 ± 0.02
DINOv3 [14] + GRAD-CAM [16]	0.65 ± 0.01	0.53 ± 0.02	0.68 ± 0.01	0.60 ± 0.05	0.54 ± 0.01
GALA-FORMER	0.68 ± 0.04	0.56 ± 0.02	0.78 ± 0.04	0.65 ± 0.01	0.58 ± 0.02

Quantitative Localization (BraTS, MSLesSeg) Dice and mean Intersection over Union (mIoU) assess voxel-level overlap. For BraTS, the three tumor subregions (enhancing tumor, tumor core, whole tumor) are merged into a binary lesion mask. Lesion-wise Recall/Precision are computed using 8-connected component analysis, and mAP@0.5 evaluates detection quality at 0.5 IoU.

Interpretability (ADNI) Faithfulness is assessed via Deletion (probability drop after removing salient regions), Insertion (probability gain after adding them), and Sensitivity-N (effect of perturbing the top- $N\%$ pixels). Neuroanatomical plausibility is assessed using region-of-interest (ROI) overlap with the hippocampus and amygdala—key medial temporal structures known to undergo early atrophy in Alzheimer’s disease—and by correlating activation strength with hippocampal volume.

5. RESULTS

5.1. Classification Performance

As shown in Table 1, DINOv3 outperforms ResNet-50 and ViT on all datasets. Improvements are largest on BraTS and MSLesSeg, where lesions are visually apparent on T_1 -weighted MRI, yielding the highest Accuracy and F1-score. Even on ADNI—where pathology is subtle and slice-level labels are inherited—DINOv3 achieves the strongest performance (AUROC = 0.84). The consistency of these gains across datasets with fundamentally different visual characteristics suggests that DINOv3 learns robust global representations that are well suited for downstream localization. These results support DINOv3 as a reliable backbone for weakly supervised anomaly detection.

5.2. Quantitative Localization on Annotated Datasets

GALA-Former achieves the best localization accuracy on BraTS and MSLesSeg (Tables 2, 3). On BraTS, it obtains Dice 0.71 and L-Rec 0.84; on MSLesSeg, it maintains strong detection for numerous small lesions (Dice 0.68, L-Rec 0.78).

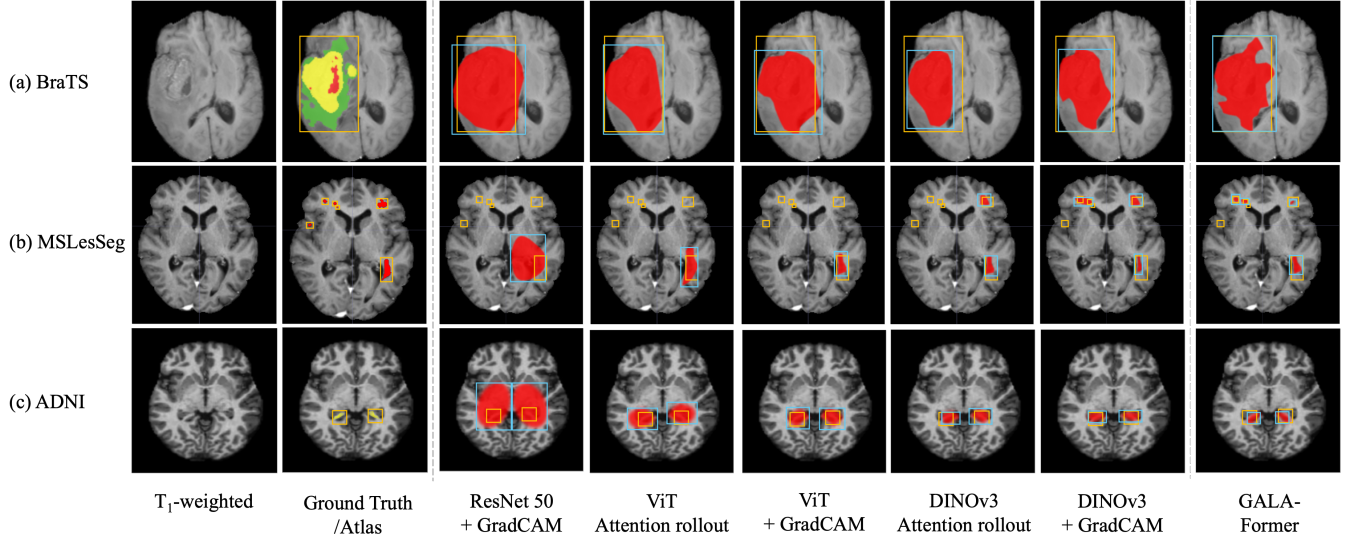


Fig. 2. Qualitative comparison of localization across baselines and GALA-Former on (a) BraTS (tumor), (b) MSLesSeg (MS lesions), and (c) ADNI (AD-related structures). Yellow boxes: ground-truth lesion masks or atlas-defined anatomical ROIs; blue boxes: predicted localization regions.

Qualitative results in Fig. 2 show that predicted blue boxes align closely with yellow ground-truth masks, whereas baselines either highlight sparse discriminative spots or produce scattered false positives. Moreover, GALA-Former preserves lesion shape better than single-cue methods, producing smoother and more contiguous activations—a critical property for clinical interpretability. These results demonstrate the benefit of combining token-, attention-, and gradient-based cues for stable and anatomically coherent localization.

5.3. Faithfulness and Anatomical Alignment on ADNI

On the unlabeled ADNI dataset, GALA-Former provides the most faithful and anatomically plausible explanations (Tables 4, 5). It achieves the lowest Deletion (0.29) and highest Insertion (0.61) and Sensitivity-N (0.20), indicating that its

Table 4. Faithfulness metrics with 95% confidence intervals on ADNI. Lower Deletion score and higher Insertion score/Sensitivity-N indicate better attribution.

METHOD	DELETION SCORE	INSERTION SCORE	SENSITIVITY-N
RESNET-50 [21] + GRAD-CAM [16]	0.41 ± 0.01	0.47 ± 0.05	0.33 ± 0.02
ViT [7] ATTENTION ROLLOUT	0.38 ± 0.02	0.51 ± 0.01	0.28 ± 0.04
ViT [7] + GRAD-CAM [16]	0.34 ± 0.05	0.55 ± 0.02	0.24 ± 0.05
DINOv3 [14] ATTENTION ROLLOUT	0.36 ± 0.03	0.54 ± 0.01	0.26 ± 0.02
DINOv3 [14] + GRAD-CAM [16]	0.32 ± 0.01	0.57 ± 0.03	0.23 ± 0.01
GALA-FORMER	0.29 ± 0.05	0.61 ± 0.01	0.20 ± 0.03

Table 5. Anatomical consistency with AD-related structures with 95% confidence intervals on ADNI.

METHOD	HIPP. OV.	AMYG. OV.	CORR. (HIPP. VOL.)
RESNET-50 [21] + GRAD-CAM [16]	0.24 ± 0.01	0.21 ± 0.01	0.28 ± 0.02
ViT [7] ATTENTION ROLLOUT	0.31 ± 0.03	0.27 ± 0.05	0.33 ± 0.01
ViT [7] + GRAD-CAM [16]	0.36 ± 0.01	0.31 ± 0.01	0.42 ± 0.03
DINOv3 [14] ATTENTION ROLLOUT	0.35 ± 0.04	0.33 ± 0.01	0.36 ± 0.01
DINOv3 [14] + GRAD-CAM [16]	0.42 ± 0.02	0.38 ± 0.03	0.48 ± 0.04
GALA-FORMER	0.48 ± 0.05	0.44 ± 0.02	0.52 ± 0.01

highlighted regions contribute most strongly to classification. Anatomically, it yields the highest overlap with hippocampal (0.48) and amygdalar (0.44) ROIs and the strongest correlation with hippocampal volume ($r = 0.52$). As shown in Fig. 2, activations cluster around bilateral medial temporal structures, aligning with known AD biomarkers, while baselines produce noisier or misplaced responses. This robustness across both annotated and unannotated datasets underscores the capacity of multi-source cue fusion to recover clinically meaningful patterns even without voxel-level supervision.

6. CONCLUSION AND FUTURE WORK

We presented GALA-Former, a Transformer-based weakly supervised anomaly localization framework that fuses token-level activation, attention-derived, and gradient-based signals to generate spatially coherent localization maps using only global diagnostic labels. Across three T₁-weighted MRI datasets, the method outperformed weakly supervised baselines, achieving competitive localization on BraTS and MSLesSeg and faithful AD-relevant patterns on ADNI without voxel-level supervision. Despite label noise from uniform slice-level propagation, especially for subtle and spatially diffuse pathology, consistent gains in faithfulness and anatomical alignment indicate that multi-signal aggregation suppresses spurious slice-level responses and captures globally relevant disease patterns.

Two limitations remain: DenseCRF may over-smooth extremely small lesions, and slice-wise training cannot model full 3D anatomical continuity. Lesion-size-stratified analysis, 3D volumetric extensions, and cross-consortium generalization will be explored in future work.

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